

COMSOFT Legacy PC-TCS™ Communications

The legacy version of PC-TCS, COMSOFT's Telescope control system, uses one of two serial (RS-232) communications protocols to communicate to other computers. Each protocol is ASCII based and human readable. Hardware speeds supported are from 1200 to 19200 baud, 8 bits, 2 stop bits, no parity. This document describes the format and command structure of each protocol.

COMSOFT Native Protocol

The Native protocol uses a continuously broadcast 150 character telemetry stream to supply the client with control system information. This telemetry stream contains column registered data as described below.

Column	Format	Description
1	Single digit integer	Motion Status of Telescope, 0= stable, 1= Declination motion, 2= Right Ascension Motion, 3= Both axes in motion.
2	Single digit integer	Current Wobble Beam, blank= Not on a Beam, 1= at Beam 1, 2= at Beam 2.
4-12	HHMMSS.ss	Current Telescope Right Ascension.
14-22	±DDMMSS.s	Current Telescope Declination.
25-33	±HH:MM:SS	Current Telescope Local Hour Angle.
35-42	±HH:MM:SS	Current Local Sidereal Time.
44-48	±DD.d	Current Telescope Elevation in degrees.
50-55	±DDD.d	Current Telescope Azimuth in degrees. - implies west + implies east.
57-61	DD.dd	Current Secant of the Zenith distance (Secz).
63	Single character	COM1 Remote command execution code. E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized Command, C= checksum error.
64	Single character	COM2 Remote command execution code. E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized Command, C= checksum error.
65	Single character	COM3 Remote command execution code. E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized Command, C= checksum error.
66	Single character	COM4 Remote command execution code. E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized Command, C= checksum error.
67	Single character	COM5 Remote command execution code. E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized Command, C= checksum error.
68	Single character	COM6 Remote command execution code. E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized Command, C= checksum error.
69	Single character	COM7 Remote command execution code. E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized Command, C= checksum error.
70	Single character	COM8 Remote command execution code. E,e= executed, 1= Bogus Request, 2= Bogus Data, 3= Unrecognized Command, C= checksum error.
72	Letter	RA Limit. R=in Limit, blank= not in limit.
73	Letter	DEC Limit. D=in Limit, blank= not in limit.
74	Letter	Horizon Limit. H=in Limit, blank= not in limit.
75	Letter	Drive Disable. D=disabled, blank= enabled.
76-83	YYYY.YY	Display Equinox.
85-93	JJJJJJ.JJ	Julian Date.
95	Single digit integer	Current com channel number.
97	±nnnnn	Focus Position
104-109	±DDD.d	Dome Error in degrees. - implies west, + implies east.
111-120	HH:MM:SS.s	UT Time.
121-148		Reserved for expansion.
149-150	terminator	CR,LF

Up to 8 separate serial channels can be handled by PC-TCS with this protocol.

Commands are simple ASCII strings, letters are all uppercase and each command is terminated with a single Carriage Return (ASCII 13 decimal). Commands execution status for a given channel is verified by reviewing the telemetry stream in character positions 63-70 for the execution code. The current channel number of the communications is given in the stream at character position 95.

The commands accepted by PC-TCS and their descriptions are as follows . . .

ABER	OFF/ON	Turns the pointing corrections for Aberration OFF/ON.
ACCESS	sssssss	Sets the security level for keyboard access to PC-TCS (password = sssssss for desired level of access.)
AUTODOME	OFF/ON	Turns TCS automatic dome motion control OFF/ON
BIASDEC	$\pm$ nnnn.nnn	Sets the declination axis "bias rate" in nnnn.nnn arcseconds per second. The rate will not become active unless a BIASON command is given or a previous non-zero rate is already active.
BIAS	OFF/ON	Turns any active bias rates OFF/ON.
BIASRA	$\pm$ nnnn.nnn	Sets the declination axis "bias rate" in nnnn.nnn arcseconds per second. The rate will not become active unless a BIASON command is given or a previous non-zero rate is already active.
CANCEL		Stops a slew in progress with full ramp down.
CDBIAS	OFF/ON	Turns the correction for cosine declination in the bias rate OFF/ON.
CDDRIFT	OFF/ON	Turns the correction for cosine declination in the paddle drift rate OFF/ON.
CDGUIDE	OFF/ON	Turns the correction for cosine declination in the paddle guide rate OFF/ON.
CDWOB	OFF/ON	Turns the correction for cosine declination in the wobble vector OFF/ON.
CLEARDIFF		Resets the difference fields on the PC-TCS Runtime display to zero.
DECLAREINIT		Synchronizes the telescope by forcing the current position to become the same as the commanded position.

DECLAREREF		Copies the current position into the Reference registers of PC-TCS.
DECLARESTOW		Changes the Stow position of the Telescope to be the current position of the telescope.
DISEPOCH	nnnn.nnn	Changes the display epoch to be “nnnn.nnn” where “nnnn.nnn” is the date epressed as a decimal year. A value of 0.0 will force the display epoch to be equal to the current date.
DOME	OPEN/CLOSE	Opens or closes the dome.
DOMEINIT		Declares the current position of the dome to be the stow position. This position is used by the STOWDOME command.
DRIFTDEC	±nnnn.nnn	Set the paddle declination Drift rate to be nnnn.nnn arc seconds per second. A negative value has the effect of reversing the sense of the buttons.
DRIFTRA	±nnnn.nnn	Set the paddle Right Ascension Drift rate to be nnnn.nnn arc seconds per second. A negative value has the effect of reversing the sense of the buttons.
ELAZ	ee.ee ±aaa.aa	Moves to the Elevation and Azimuth specified by ee.ee and ±aaa.aa degrees respectively.
EPOCH	nnnn.nnn	Changes the input coordinate epoch to be “nnnn.nnn” where “nnnn.nnn” is the date epressed as a decimal year. A value of 0.0 will force the input epoch to be equal to the current date. The input epoch is used to precess an entered coordinate to true of date coordinates.
FILEPERM	filename	Sets the source input mode to PCF catalog mode using the PCF file specified by “filename.” (“filename” should not contain the “.PCF” extension.)
FILEUSER	filename	Sets the source input mode to USER catalog mode using the USER file specified by “filename.” (“filename” should not contain the “.CAT” extension.)
FLEX	OFF/ON	Turns the correction for flexure OFF/ON when processing new input coordinates.
FLEXRATE	OFF/ON	Turns the correction rates for flexure OFF/ON.
FOCUS	±nnnn	Moves the focus motor “nnnn” steps.
GUIDEDEC	±nnnn.nnn	Set the paddle Declination Guide rate to be nnnn.nnn arc seconds per second. A negative value has the effect of reversing the sense of the buttons.

GUIDERA±nnnn.nnn

Set the paddle Right Ascension Guide rate to be nnnn.nnn arc seconds per second. A negative value has the effect of reversing the sense of the buttons.

KILL

Disables the PC-TCS drive system. This command acts like an “Emergency Stop” command and should be used only in the most severe of circumstances.

LIMITENABLE

Enables limit detection and processing in PC-TCS.

LIMITOVERRIDE

Disables PC-TCS limit detection and processing.

MANUAL

Sets the Source Mode of PC-TCS to Manual - (keyboard like) where RA and DEC's must be individually specified rather than retrieved from a file. This mode command must be given before a NEXTRA and NEXTDEC command can be executed.

MIRROR

OPEN/CLOSE

Opens or closes the mirror covers.

MOVB1

Moves to the current position plus the value of the Wobble Vector.

MOVB2

Moves to the current position minus the value of the Wobble Vector.

MOVDOME

aaa.aa

Moves Dome to the azimuth aaa.aa.

MOVNEXT

Moves to the position stored in the NEXT Registers of PC-TCS.

MOVOFF

Moves to the current position plus the value of the Offset Vector.

MOVREF

Moves to the position specified in the Reference Register of PC-TCS.

MOVSTOW

Moves to the current Stow position of the telescope. This Command also disables tracking.

MTIME n

Sets the focus speed - larger numbers = slower speed.

NEXTDEC

±ddmmss.s

Sets the Next Declination Register in PC-TCS to the value “ddmmss.s” where dd = degrees, mm= minutes, and ss.s= seconds. The MANUAL command should be executed before this command is given.

NEXTRA ±hhmmss.s

Sets the Next Right Ascension Register in PC-TCS to the value “hhmmss.s” where hh = hours, mm= minutes, and ss.s= seconds. The MANUAL command should be executed before this command is given.

NOP

A “No Operation” Command - does nothing. Used to test remote communications.

NUTAT OFF/ON

Turns the pointing corrections for nutation OFF/ON.

OBJECT nnnnn

Retrieves and processes the coordinates in record number “nnnnn” in the catalog file selected using the FILEPERM or the FILEUSER modes. The processed coordinates are then stored in the NEXT registers of PC-TCS. A MOVENEXT command will then move the telescope to these coordinates.

OFFDEC ±ddmmss.s

Sets the Declination component of the Offset Vector to “ddmmss.ss” where dd= degrees, mm= minutes, and ss.ss= seconds.

OFFRA ±hhmmss.s

Sets the Right Ascension component of the Offset Vector to “hhmmss.ss” where hh= hours, mm= minutes, and ss.ss= seconds.

PARA OFF/ON

Turns the pointing corrections for parallax OFF/ON.

PREC OFF/ON

Turns the pointing corrections for precession OFF/ON.

PROPER OFF/ON

Turns the pointing corrections for proper motion OFF/ON.

RADECGUIDE ±RR.rr ±DD.dd

offsets the current position of the telescope by RR.rr arcseconds in RA and DD.dd arcseconds in DEC. Used for remote guiding (autoguiding)

RECALLPOS nnn

Recalls the saved position (see SAVEPOS command) and copies the coordinates to the NEXT registers of PC-TCS. The telescope can be moved to this position using the MOVENEXT command.

REFRAC OFF/ON

Turns the pointing corrections for refraction OFF/ON.

REFRRATE OFF/ON

Turns the rate corrections for refraction OFF/ON.

REMOTECHK OFF/ON

Turns serial command checksumming OFF/ON.

SAVEPOS nnn

Saves the current position to the Save register number “nnn”. Use the RECALLPOS command to retrieve the value.

SETDATE mm/dd/yyyy (remote only)
 Sets the UT date of PC-TCS to “mm/dd/yyyy” where mm= month, dd= day, yyyy= year.

SETTIME hhmmss.ss (remote only)
 Sets the PC-TCS system UT time to “hhmmss.ss” where hh= hours, mm= minutes, ss.ss= seconds.

SLEW OFF/ON (deadman slew systems only)
 Turns OFF/ON the slew capability of PC-TCS (speed limited to @ 1/5 degree per second).

STOWDOME
 Moves the dome to the Stow position as specified in the PFEDIT program. Note: the stow position is the position the dome is at when TCS is started or the position the dome is at when the Declare-Dome Stow menu item is activated or the DOMEINIT command is given.

TRACK OFF/ON
 Turns OFF/ON sidereal tracking and correction rates.

UNABER OFF/ON
 Turns OFF/ON the deconvolution for Aberration on the RA and DEC display

UNFLEX OFF/ON
 Turns OFF/ON the deconvolution for Flexure on the RA and DEC display

UNKILL
 Enables PC-TCS drive capabilities. (Used after a previous KILL command).

UNREFR OFF/ON
 Turns OFF/ON the de-convolution for Refraction on the RA and DEC display

WOBDEC ±ddmmss.s
 Sets the Declination component of the Wobble Vector to “ddmmss.ss” where dd= degrees, mm= minutes, and ss.ss= seconds.

WOBRA ±hhmmss.s
 Sets the Right Ascension component of the Wobble Vector to “hhmmss.ss” where hh= hours, mm= minutes, and ss.ss= seconds.

ACL PROTOCOL

The Astronomical Control Language (ACL) is a Query/Response protocol. This protocol is supported by ASCOM - a standardization movement used by many commercial vendors.

All commands and responses are ASCII strings terminated by a single Carriage Return (ASCII 13 decimal). Case is important in this protocol and most commands are lowercase. Currently PC-TCS can support only a single channel (one RS-232 communications port) using this protocol.

The commands and requests PC-TCS supports are as follows . . .

Command	Parameter(s)	Response	Description
cr		ok	Acts like operator typing a single Carriage return key on console
device		ok acl PC-TCS X.xx.x	Returns PC-TCS version.
dome	open	ok	Opens dome shutter
	close	ok	Closes dome shutter
	on	ok	Enables auto-dome azimuth control
	off	ok	Disables auto-dome azimuth control
	init	ok	Forces PC-TCS to believe dome is currently at its predefined stow position
	getaz	ok XXXX	Returns dome azimuth in tenth's of a degree
	setaz XXXX	ok	Moves dome to an azimuth of XXXX/10 degrees
	home	ok	Moves dome to its hardware defined stow position
	stow	ok	Moves dome to its predefined software stow position
	status	ok MDLSAAAAH	M= mode, 0= off, 1= auto, 2 = stow D= direction 1= right, 2= left, 0= stationary S= shutter (o= open, c= close) L= lower shutter (o= open, c= close) AAAA= dome azimuth in degrees H= position (h= at hardstow, p= on position)
esc		ok	Acts like operator typing a single ESC key on console
F+		ok	Moves focus positive number of steps set previously by FS or FF command
F-		ok	Moves focus negative number of steps set previously by FS or FF command
FF		ok	Sets fast focus mode for TheSky
FQ		ok XXXXXXXX	Returns current focus position
FS		ok	Sets slow focus mode for TheSky
focus	±XXXXXX	ok	Moves focus to absolute position XXXXXXX
	?	ok focus= ±XXXXXX	Returns focus position in counts
	status	ok m ±XXXXXX	where m= 0 if stable, 1 if moving, XXXXXXX is current position.
	setlimit HHHHH LLLLLL	ok	HHHH is high limit value and LLLL is low limit value.
	getlimit	ok HHHHHH LLLLLL	Returns focus limits
get	focuspos	ok focus= ±XXXXXXXX	Returns focus position in counts
	ra	ok XXh XXm xx.xxs	Returns RA position
	dec	ok XXdXXmXX.xxs	Returns DEC position
	altitude	ok ±XXX.xx	Returns telescope altitude in degrees
	azimuth	ok ±XXX.xx	Returns telescope azimuth in degrees (0= north, 90= east)
	elevation	ok ±XXXXX.xx	Site elevation above sealevel in meters
	utd	ok mm/dd/yyyy	System UT date
	utc	ok hh:mm:ss	System UTC time
	telescopeName	ok name	Returns name of telescope
	telescopeinfo	ok	Reserved for expansion
	epoch	ok XXXX.xxxx	Returns display epoch (equinox)
	decrate	ok ±XXXX.xx	Dec bias rate in arcseconds per second
	rarate	ok ±XXXX.xx	RA bias rate in arcseconds per second

	flag n	ok X	Telescope motion status. If n= 0 - returns 1 if tracking; if n=1 - returns 1 if telescope or dome is moving. All other cases returns 0.
	sidereal	ok xxhxxmxx.xxs	Returns sidereal time
	latitude	ok ±XX.xxxxxx	degrees NORTH
	longitude	ok ±XXX.xxxxxx	degrees EAST
	find	ok	Moves telescope and dome to stow positions
	setup	ok	Sets stow position for telescope to be current DEC and Hour Angle
	x_parkset	ok	Sets stow position for telescope to be current DEC and Hour Angle.
goto	objectra XXhXXmXX.xxs objectdec ±XXdXXmXX.xxs	ok	Sets next position RA & DEC
		ok	Moves to Next Position RA and DEC
	objectalt ±XX.xxx objectaz XXX.xx	ok	Moves to Alt & Az
home	find		Moves telescope and dome to stow positions
	setup		Sets stow position for telescope to be current DEC and Hour Angle
	x_parkset		Sets stow position for telescope to be current DEC and Hour Angle.
limits	?	ok LLLLL	limit status: LLLLL= NONE RA_LIMIT violation DEC_LIMIT violation HOR_LIMIT violation SECZ_LIMIT violation HARDWARE violation
	inhibit	ok inhibit!	Turns all limit detection off
	enable	ok	Turns limit detection on.
move	ra ±XXXX.xx	ok	where XXX.xx is in arcseconds.
	dec ±XXXX.xx	ok	where XXX.xx is in arcseconds.
script	on	ok	Starts previously loaded script execution
	off	ok	Stops previously loaded script execution.
	NNNNNNNN	ok	Load script file NNNNNNNN.tcs Note: the .TCS extension is implied.
set	objectra XXhXXmXX.xxs objectdec ±XXdXXmXX.xxs	ok ok	Stores passed RA into PC-TCS “next position” registers
	decrate ±XXXX.xx.	ok	Set and activate Dec bias rate of XXXX.xx arcseconds per second.
	rarate ±XXXX.xx	ok	Set and activate RA bias rate of XXXX.xx arcseconds per second.
	utd mm/dd/yyyy	ok	Set UT date
	utc hh,mm,ss.ss	ok	Set UTC
	epoch XXXX.xx	ok	Set precesssional equinox of Next Object RA and DEC coordinates to XXXX.xx send this information before sending goto or set ra dec commands.
	longitude ±XXX.xxxx	ok	Set system longitude to XXX.xx degrees west.
	latitude ±XX.xxx	ok	Set system geodetic latitude in degrees north.
	elevation ±XXXX.xx	ok	Set system elevation in meters above sealevel.
	focuspos ±XXXXXX	ok	Move focus to XXXXX counts absolute.
	flag VV (x)		Where: VV= 0 - if x= 0 turn tracking off VV= 0 – if x=1 turn trakin on VV= 16 – toggle flexure rates on/off VV= 23 – toggle PEC rate on/off
shutter	open	ok	Open dome shutter
	close	ok	Close dome shutter
	status	ok m{o,c} l{o,c}	m= main shutter l= lower dropout shutter o= open, c= close
stop		ok	Motion cancel (same as F9 key in PC-TCS)
status		ok dddddddd	Returns acl status (for TheSky)

sync		ok	Synchronize telescope coordinate system to match previously sent goto or set objectra and objectdec coordinates.
x_parkset		ok	Null operator (used to satisfy a bug in TheSky)

Errors will return "error XX" instead of the "ok" status. The XX will have a value between 1 and 30 which indicates either bogus data (XX= 1-12, 14-30) or unknown command (XX= 13).

PCTCS-NG Network Protocol

The next generation of control system being implemented by COMSOFT (PCTCS-NG) has only a TCP/IP network interface. This new control system will be an imbedded headless system with the only interface to the user being a multi-socketed TCP/IP connection.

To make Legacy PC-TCS somewhat compatible with this new interface, COMSOFT has a program called TELCOM which acts as an NG protocol server for Legacy PCTCS. To provide a network presence to Legacy PCTCS, the TELCOM software must run on a separate computer connected serially to the Legacy PCTCS machine and also be connected to the network with a static IP.

Communications with the TELCOM server are started by connecting to port 5750 at the host IP. Communications with the server are in the form of either a REQUEST or COMMAND. REQUESTs return information from PCTCS and COMMANDs give PCTCS directives. Both COMMANDs and REQUESTs are uppercase ASCII and have a common packet wrapper as follows . . .

PCTCS-NG Packet Structure		
Field	Description	Details
TELID	Telescope ID	The Telescope Control system Moniker - typically 4 to five uppercase characters - no whitespace - eg. KMTN1, KMTN2
SYSID	Subsystem ID	The subsystem ID uppercase string - no whitespace - eg. TCS, DOME, FOCUS
PID	Packet ID	A sequence number for the packet - set this to an ASCII representation of a number between 0 and 255
Command/ Request	COMMAND or REQUEST	Uppercase ASCII string containing the actual command appropriate to the subsystem.

Currently only the NATIVE protocol of Legacy PCTCS is supported in the TELCOM server software. Legacy PCTCS COMMANDS are identical to the commands documented in the serial communication for the NATIVE protocol communication.

Legacy PCTCS REQUESTs are as follows . . .

Request keyword	Response	Description
ALL	Telemetry stream	Returns entire 150 character telemetry stream string as documented above.
HA	±HH:MM:SS	Local Hour Angle of Telescope
RA	HHMMSS.ss	Current RA of Telescope
DEC	±DDMMSS.s	Current Declination of Telescope
ST	±HH:MM:SS	Local Sidereal Time
EL	±DD.d	Current Elevation of Telescope
AZ	±DDD.d	Current Azimuth of Telescope
SECZ	DD.dd	Current Secant of the Zenith distance of telescope
EQ	XXXX.xx	Display epoch
JD	JJJJJJ.jj	Julian date
BEAM	N	Current Beam of telescope
MOTION	N	Motion Status of Telescope, 0= stable, 1= Declination motion, 2= Right Ascension Motion, 3= Both axes in motion.
FOCUS	±nnnnn	Current focus position
DOME	±DDD.d	Current dome error in degrees.

As examples of fully constructed COMMAND and REQUEST packet strings . . .

A request for the current Telescope Right Ascension:

```
KMTN1 TCS 1 REQUEST RA←
```

would return . . .

```
123456.78←
```

A command for the telescope offset 10 arcseconds east in RA and 4 arcseconds south in DEC:

```
KMTN1 TCS 2 COMMAND RADECGUIDE 10.0 -4.0←
```

would return a response of:

```
OK←
```

In the above examples the symbol "←" is a carriage return (ASCII 13 decimal).

All fields in both the packet structure, command and request string should be separated by one or more spaces (ASCII 20 hex).

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